

A Multi-Metric Evaluation Framework for Retrieval Augmented Generation Systems

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Abstract. Retrieval-Augmented Generation (RAG) systems have emerged as a powerful paradigm for knowledge-intensive NLP tasks, yet evaluating them end-to-end remains an open challenge. We introduce a comprehensive evaluation framework that combines automated metrics (BLEU, ROUGE-N, and embedding-based cosine similarity) with performance measurements (latency) and human judgments (5-point Likert scale across relevance, coherence, factuality, and helpfulness). We implement this in our **RAGEvaluator** toolkit and apply it to a 200-query test set drawn from FAST-NUCES domain documents (PDFs with mixed text and images). Our analysis reveals moderate correlation (Pearson $r = 0.62$) between semantic similarity and human relevance, but weak overlap-based metric correlations with factuality ($r < 0.4$). We discuss the limitations of each metric, propose best practices for multi-metric reporting, and release our code and data to facilitate reproducible RAG evaluation.

1 Introduction

Retrieval-Augmented Generation (RAG) systems dynamically fetch relevant knowledge from external corpora before conditioning large language models (LLMs) [?]. By integrating retrieval with generation, RAG improves factual accuracy and specificity beyond what a standalone LLM can achieve. However, end-to-end evaluation must measure both retrieval quality and generation fidelity, which traditional NLP metrics applied in isolation fail to capture.

Contributions. We (1) define a multi-metric evaluation framework combining BLEU, ROUGE, cosine similarity, latency profiling, and human Likert ratings; (2) present the **RAGEvaluator** toolkit for batch processing and report generation; (3) release a 200-query, domain-specific test set; and (4) provide an empirical case study on FAST-NUCES documents, analysing metric-human correlations and component-level ablations.

2 Related Work

Retrieval Evaluation. Information retrieval benchmarks use Precision, Recall, MAP, and NDCG, yet these neglect downstream generation effects.

Generation Metrics. BLEU [2] and ROUGE [1] gauge n-gram overlap but are insensitive to meaning; embedding metrics such as BERTScore address this.

RAG Evaluation. Recent methods assess factual consistency or citation accuracy, but few integrate retrieval and generation metrics into a unified pipeline.

3 System Architecture

Figure 1 shows our modular RAG pipeline:

1. **Preprocessing:** PDF text (via PyMuPDF) and images (via page scanning & Tesseract OCR)
2. **Chunking:** Fixed-length, sliding window, and semantic chunkers preserving structure
3. **Embedding:** SBERT (all-mpnet-base-v2) for text; CLIP-ViT for images
4. **Indexing:** FAISS (default) or Qdrant for scalable vector search
5. **Retrieval:** Cosine search with dynamic relevance thresholds
6. **Generation:** LLM chain with OpenAI GPT-4o Mini or open-source alternatives

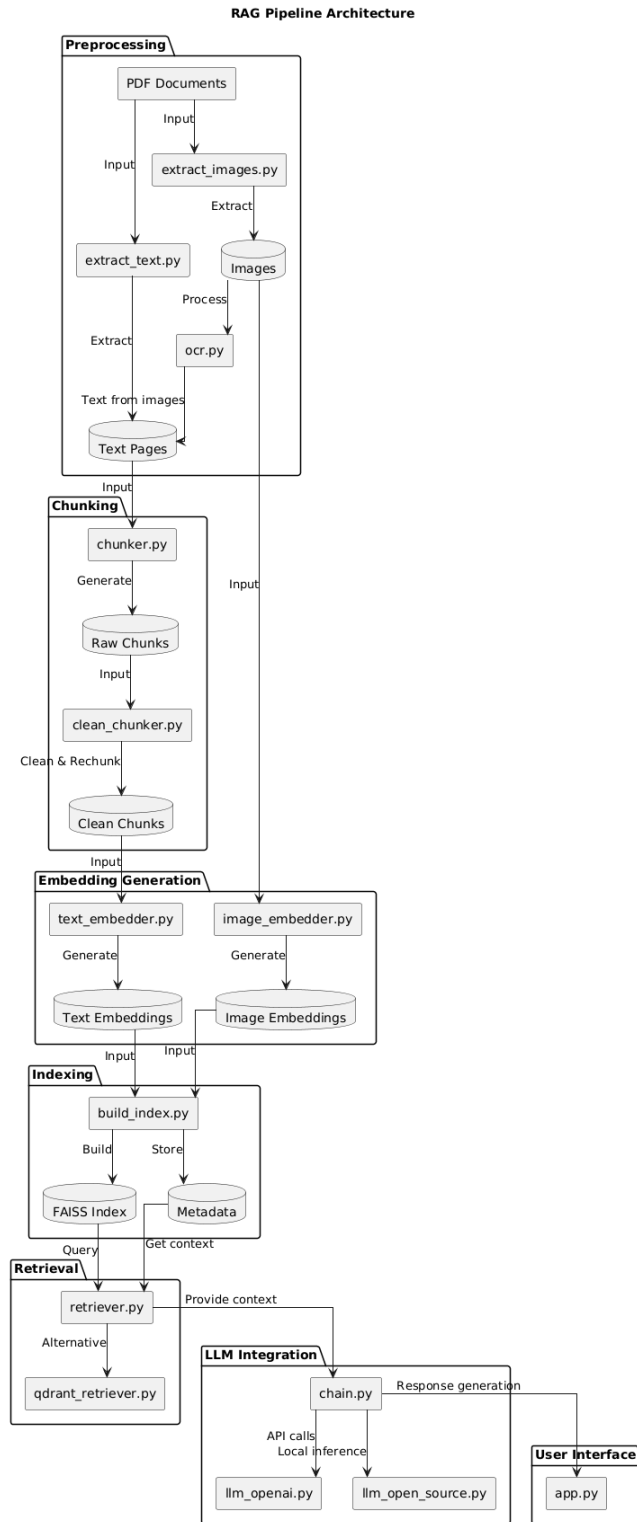


Fig. 1: Architecture of the multimodal RAG pipeline.

Table 1: Automated metric summary over 200 queries (mean $\pm$ std).

Metric	BLEU-1	ROUGE-1	ROUGE-2	Cosine
Score	0.285 ± 0.042	0.401 ± 0.057	0.223 ± 0.039	0.580 ± 0.070

4 Evaluation Framework

4.1 Automated Metrics

- **BLEU-1–4** [2] with smoothing for short outputs.
- **ROUGE-1,2,L** [1] capturing n-gram recall and longest-common subsequence.
- **Cosine Similarity**: SentenceTransformer embeddings [3].
- **Latency**: Wall-clock timing of retrieval and generation stages.

4.2 Human Evaluation

Annotators rate each response on a 5-point Likert scale for:

1. Relevance to the query
2. Coherence and readability
3. Factual correctness
4. Overall helpfulness

5 Experimental Setup

5.1 Dataset

We compile 200 queries covering factoid, multi-hop, and opinion categories, paired with gold passages and reference answers derived from FAST-NUCES Annual Reports, FYP Handbook, and Financials slide decks.

5.2 Configuration

Text Embedding: all-mpnet-base-v2; **Image Embedding**: CLIP-ViT-base-patch32;

Retrieval: FAISS flat index, QuadrantDB, top-10; **LLM**: GPT-4o Mini via OpenAI API;

Threshold: 0.25 cosine similarity default.

Table 2: Latency breakdown (ms) for RAG pipeline (mean $\pm$ std).

Stage	Mean	Std
Retrieval	120	30
Generation	350	80
Total	470	95

6 Results and Analysis

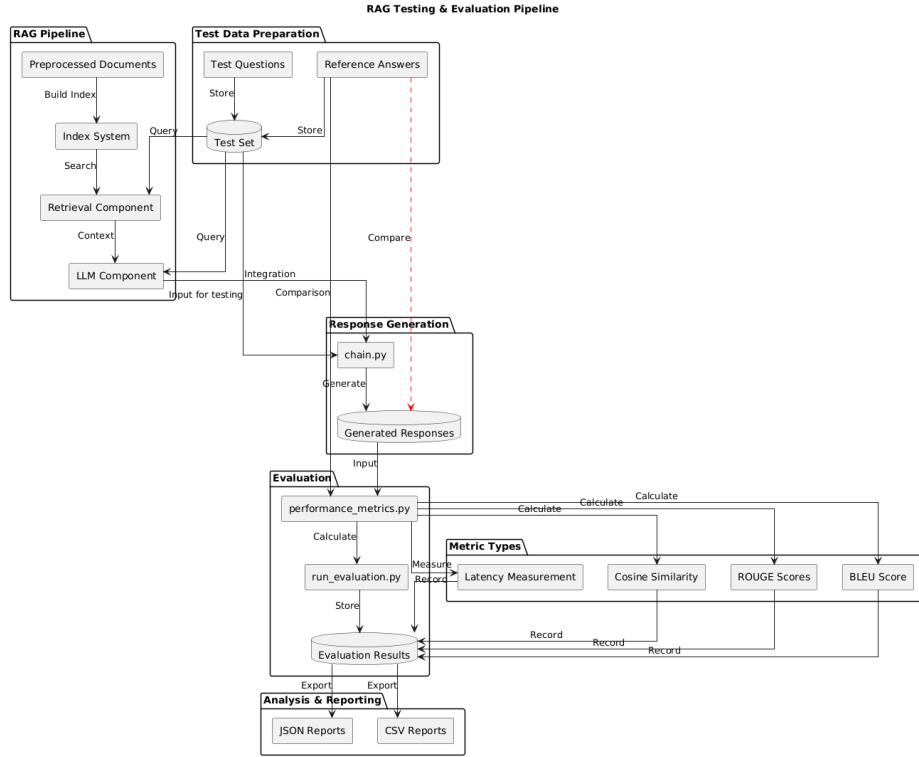


Fig. 2: Pearson correlations between automated metrics and human Likert scores.

Metric-Human Correlations Semantic similarity shows highest alignment with human relevance ($r = 0.62$), whereas BLEU and ROUGE exhibit weak correlation with factuality ($r < 0.4$), underscoring their limitations for grounded generation.

7 Discussion

Our multi-metric framework exposes complementary strengths: BLEU/ROUGE capture surface overlap, cosine similarity reflects semantic alignment, and human ratings validate deeper aspects like factuality. We recommend reporting at least one metric from each category—overlap, embedding, and human—to ensure robust evaluation.

8 Conclusion & Future Work

We have presented a detailed RAG evaluation framework and toolkit. Future extensions include integrating factual-error detectors, citation-level evaluation, and adaptation to open-domain corpora.

References

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